Biology Becomes Predictive

A GBN Interview with Robert Carlson
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Scientist Robert Carlson describes how our understanding of biology is becoming more like physics and enabling us to build living things with confidence.

GBN Global Business Network

Robert Carlson is a physicist-turned-biologist. This is actually an important distinction given the theoretical and sociological differences that distinguish these fields. Physics has tended to focus on reducing our understanding of the physical world to its simplest components and discovering the fundamental laws that govern everything. As a result, the field has tended to attract people who like clarity as well as engineering types who can build things based on those clear rules. Biology, on the other hand, has tended to be a much messier field that resists reductionism. The complexity of living things forces biologists to be content with describing rather than definitively understanding the underlying biological processes. They must be comfortable watching, rather than building, things—at least up till now.

Carlson, who describes himself as someone who likes to build things, has crossed over to biology partly because the field is becoming more like physics as it continues to become more predictive (in a scientific vs. a futuristic sense). Physics is predictive because scientists know enough about its fundamental laws and have so refined the math that they can predict with certainty what will happen when, for example, a ball is dropped from a building. Biologists are just beginning to say with some certainty what happens when a certain gene is turned on or off. Now, biology is moving so fast that the predictive abilities are mounting and this has huge ramifications for both the field and the world at large.

Carlson is part of a wave of young scientists who are bringing something of a hacker mentality to the life sciences. The tools of biology and biotech are becoming cheaper and more democratized. When I interviewed Carlson at the Molecular Sciences Institute in Berkeley, he showed me the bench tools that are considered cutting edge. He anticipates that such tools will be widely available within a few years for people to use in their garages. Carlson has since moved to Applied Minds, a research and invention laboratory in Southern California, cofounded by Danny Hillis, where he helps build technological prototypes for companies and organizations who outsource their R&D.

As Carlson says in the interview, this new ability to build things can have some negative consequences as well. Right now we 're learning enough to create potentially harmful organisms without knowing enough to neutralize those organisms if they get loose or are used as weapons. Carlson estimates that we have about a ten-year window of vulnerability where the "bad guys" might have the advantage over the good guys. He argues, however, that the only way forward is to rapidly close that knowledge gap and accelerate the development of a more comprehensive understanding of biology. Then, when a potentially lethal virus is unleashed, on purpose or by accident, our society could respond immediately by understanding how it works and how it can be controlled. If biology becomes truly predictive, then we will be able to dismantle things as easily as we build them. We can't do either now, but thanks to the efforts of Carlson and the younger generation of scientists like him, we may be able to do both much sooner than expected.

—Peter Leyden

Topics Covered in this Interview

Meeting Sydney Brenner

Peter Leyden (GBN): GBN is a network of businesspeople, governments, and organizations from all over the world. And this project involves connecting with interesting people in different fields and finding out about what they see happening in the next 10 years. Let's begin with your story, which is an interesting one because it starts not with biology but–

Robert Carlson: A train ride.

GBN: How does it start with a train ride?

Carlson: In 1996 I was at Princeton, working in physics and biology. My background is all physics, but I've always done biology within that context. I was on a train from Princeton to New York and met a guy whom I just happened to see on campus earlier that day, and we started chatting. I had no idea who he was. He seemed interested in what I was doing, so I told him my story, what I was working on and how I had this crazy scheme to try to find out how smart single cells are. That's what I was interested in at the time, thinking about what kinds of computations a single cell can do. That's still the way I frame the kind of problems I'm trying to solve, although what I'm working on right now doesn't necessarily relate to that directly.

So I told him the story. He seemed interested and said, "Well, you should work for me." I was already in my Manhattan frame of mind: take everything with a grain of salt. He said, "My name is Dr. Sydney Brenner."

I thought, "OK, Dr. Sydney Brenner, a job, that's fine." Then I got to Manhattan and went to the NYU med school library to do some research. I headed back to campus later that day and found out that Sydney Brenner is a big guy—one molecular biology professor said, "Sydney Brenner! He's like a god to me!" It was a real job. One thing led to another. Sydney was trying to figure out where to put the Molecular Sciences Institute at that time.

GBN: Tell us about Dr. Brenner.

Carlson: Sydney Brenner is a grand old man in molecular biology. Maybe not the grandfather, but perhaps the grand-uncle of molecular biology. He was at Cambridge in the early days, working with Francis Crick et al. He was in on the elucidation of mRNA, or messenger RNA, and the triplet code. Over the years he also started the *C. elegans* project with Francis Crick, deciding that the worm was a good model organism to work on, in that you could watch all of the cells, you could see the organism develop, and that might be a good way to study development and neurobiology. Later on he started the project to sequence Fugu, which is the puffer fish. That's an interesting model for vertebrates in particular because it has very little "junk DNA," as they say. It has a small genome for the number of genes that it has compared to other vertebrates.

So that's the kind of person Sydney is. He starts interesting things and then goes away and starts another interesting thing. He's a serial project initiator.

GBN: So what had he started then?

Carlson: He was in the process of starting the Molecular Sciences Institute. He had some money and he said to me, “I have this place that I’m starting, and the idea is to do the science that no one else is willing to do—to do the crazy stuff, because I think there isn’t enough creativity in science. It’s too governed by, at least in molecular biology, what peer review committees say or what grant review committees say. There isn’t enough new stuff out there.”

So he said, “Come do the science no one else is willing to do.”

And I said, “OK! I’m all for that.” He landed the Institute here in Berkeley and I was looking to be in San Francisco, so it worked out nicely.

GBN: What year was this?

Carlson: I joined in the summer of’97.

GBN: How would you characterize what they do?

Carlson: Sydney’s perspective was, as I said, to do the science no one else was willing to do. Not use kits, for example. A lot of sequencing, and molecular biology in general, at this point happens using kits. You get a box; there are some solutions in the box with bottles that have colored labels, there are some instructions. The instructions say, “Take your sample, put it in a tube, add 100 microliters of solution A and 100 microliters solution B,” etc., which is very useful but isn’t very creative. You’re not solving new problems, necessarily. What Sydney saw was that a lot of people were just using these tools to do the same old thing that they’d always done but in a faster way rather than a new way.

Roger Brent, who is the director now, had the attitude: “Well, the genome will be sequenced soon, so then what will we do? Molecular biology in the last few years has become very focused on understanding the genome, but what does that really tell us? What will we do when we have that information?”

From that, the goal of the Institute has evolved into trying to bring about what we’re calling a predictive biology. We’re trying to make biology more like physics, in the sense that we’re hoping to create some quantitative predictive models in biology, and also create some tools that allow testing of those models at the molecular level.

“Storytelling” Differences Between Biology and Physics

GBN: When you say the word predictive, it has a lot of associations. What do you mean in a scientific sense?

Carlson: The real difference is the word quantitative. It’s the difference in the way stories are told in physics and stories are told in biology. When stories are told in biology, it tends to be done with natural language. I go out in the field and I see this organism behaves in a certain way. I see that worms come up when the ground gets wet and they go underground when they can, when they are not going to drown. They eat particular things. They behave in particular ways.

That kind of story tends to be the way biologists describe molecular events as well. Here’s a protein that is turned on under certain circumstances. I know that this protein effectively is a wrench and it adds a nut onto this other protein, which is a bolt.

Those are the kinds of stories you tell, but they’re not quantitatively predictive stories. It’s not like you say, “I dump a hundred molecules of molecule X into the environment around the cell and I know that cell responds by making a thousand molecules of this something else.” There have been very few models like that in molecular biology until very recently.

GBN: What would be the physics way of telling the story?

Carlson: The physics model is an equation, to some extent. Of course there is language that goes along with defining equations. There’s an intuition that goes along with finding the equation, but what you’re trying to do is to make a prediction—that is, how much of one thing happens when I twiddle the knob that controls the system by a certain amount—in a quantitative way. I have a ball in my hand. I drop it. I can predict fairly well how high it will bounce as a result of that. It’s not a natural language description that provides me that power, it’s calculus, and it’s some understanding of the fundamental physical universe.

Biologists tend not to describe life in that way. The point of the Institute is to try to create something more like the quantitative description that occurs in physics. That might occur in the following way. We can relatively quickly—if there are enough graduate students around—determine the genetic sequence of any organism we choose. Now we have the genome of the organism; it’s a set of genes arrayed in a certain way in the DNA, and at this point in time you can monitor when those genes are turned on and off by looking at a process called transcription, where the genetic code in DNA gets turned into messenger RNA, or mRNA. The tools are also coming along to measure the next step, which is finding out how much of that mRNA gets translated into another code, this time for proteins. We can apply these measurement techniques to describe how a particular signal transduction pathway works, for example. A signal transduction pathway converts some information about the environment around a cell – the concentration of a certain molecule, or the temperature, for example – into a molecular signal the cell uses to decide what to do next.

You start by measuring the transcripts, the transcription events that occur, to find out when genes get turned on and off. Then you can also measure very carefully, if you're good and if you're lucky, how the amounts of proteins that correspond to those transcripts change in time and also how the proteins change their behavior in time—how they bind to different things, or where they go in the cell and when they do that. Out of that kind of information, you could hope to build a quantitative model that's predictive in the physical sense, so that you could predict the behavior of the system in circumstances that are different than you have previously measured.

GBN: I see.

Carlson: I have a ball. I throw it and I watch how it bounces. Say I know something about the ball's mass and, depending on how much physics you want to include, the shape and how smooth it is and that sort of thing. I can predict where the ball is going to go before I throw it because I know enough about the ball and the way it interacts with the world around it to be able to predict the trajectory.

It would be nice, for a variety of reasons, to be able to do that in biology. One reason is that such an understanding will probably—it's not completely clear—lead to a better description of how drugs work, for example, or how disease works. For example, one good version of a more quantitative story in biology concerns the change in the gene for hemoglobin that results in sickle-cell anemia. The reason that sickle-cell anemia is still around is that it confers some resistance to malaria. It also, in the more severe forms, is an incredibly painful and debilitating disease. Science has been trying to tell that story for some decades now. It's to the point where people understand not only the particular place in the genome—the particular single nucleotide polymorphism (or SNP), the one base pair change that happens between normal hemoglobin and sickle-cell hemoglobin—but also the thermodynamics that describes how sickle cell hemoglobin polymerizes, or aggregates simple molecules to form more complex molecules with different properties, and how that alters the interaction with the pathogen that causes malaria. That story is very detailed and very quantitative from an empirical point of view, but it's not a predictive story in that you can't take the information about how the sequence of the gene has changed and quantitatively predict how the SNP changes the structure of the protein, or how the change in structure influences the behavior of the protein.

GBN: So the distinction between the two kinds of sciences is not necessarily empirical, which is watching, seeing, and measuring what happened. It's actually going deeper to understand the mechanisms that make it happen. For example, if you poke it here you know exactly what will happen, not because you saw it before, but because you understand what would happen in the chain of events between that action and the result.

Carlson: That's exactly right. Here's another example of the inability at the moment to tell an even simpler story about biology, but one that's along the same vein.

Say you have a genome for some organism and you can sequence that genome very quickly, so you find out what genes are there. You can go look in a database and see which genes that we already know about are like the ones in the organism that you just found. Well, even if you have that information, you still really can't say whether the organism is going to have two legs or four legs or whether it's going to fly or whether it's going to slither. That general kind of story, the ability to read the genome and understand how the pieces are going to interact and what that organism is going to develop into, would be useful from a scientific standpoint.

The ability to make quantitative predictions would also be useful when we think about design in biology. We already use biology as technology – we have for thousands of years as we've farmed and bred animals, etc. The manipulation of biology, of human interaction with the environment, just through the use of antibiotics is an extraordinary example in itself of using biology as technology. Penicillin is a biological technology. But these are not examples of "engineering" in the standard sense.

Biological engineering doesn't currently exist in the same way that, say, electrical or aeronautical engineering does. For one thing, there is little or no basis for risk analysis at the moment. The current flap about genetically modified foods, for example, is really more about the fact that we don't know what we're doing rather than there being something intrinsically bad about interfering with biology. If there were something intrinsically bad about interfering with biology, by that logic we ought not to be using antibiotics. Antibiotics, along with basic sanitation, basically by themselves have probably expanded the human life span by nearly a factor of two in the last hundred and fifty years.

The ability to better engineer biological systems – to better design biological systems – will affect the behavior of that biological system down the road. Quantitative prediction will not only be useful in the implementation of real biological engineering, but also in allaying fears that the organism that results will be detrimental.

GBN: That's a good distinction to make. It sounds like the worries are not so much with meddling with biology, but knowing what we're doing when we're meddling with it.

Carlson: We've been meddling for a long time. When people like Jeremy Rifkin acknowledge that we've been meddling with biology for a long time, he's astute enough to make the point that the timescale over which we're doing it now is very different than in the past. The ability, and this is my addition to this argument, for us to respond to any changes we might make is nil, effectively, at this point. We can't do anything about any changes we make that might be detrimental.

GBN: I do want to talk about that, but let me close the loop on the comparison of physics and biology. There are two ways a lay person could understand this. They could say that physics is easy to figure out because we figured it out earlier, and biology is far more complex so it's just going to take us a lot longer to figure it out. The other way to say it is that we're at a different point in the development of that science. In other words, physics is really mature and we're building on centuries of work, while biology is a relatively young science and we're just starting to get our minds around it. How would you frame it? Why is biology so different from physics in terms of our understanding that predictability?

Carlson: Well, it does get difficult to compare biology and physics in many ways. I'll start off with the following observation. A lot of people said that when we had the genome sequenced we were basically doing the quantum mechanics of biology. That just sounds wrong to me, because quantum mechanics is all about developing predictive models. All the genome really provides is a parts list, and it's not even that developed because the genome is an incomplete list of parts. Beyond that, how things interact is more important in biology than just the things that are there. The genome tells us very little, if anything at all, about how things interact.

One of the reasons that biology is making some strides now, the reason you see all these multidisciplinary efforts springing up all over the place between physics, biology, engineering, and computer science, is that the tools to measure biology have finally reached the point that you can begin to consider building predictive models. A lot of people are thinking about building predictive models of biology, meaning a model that is a quantitative description of the system. The implementation of the model might be a simulation, the testing of that model computationally, or perhaps just working out the consequences of the model on paper. The data has only just started to arrive that allows careful modeling at the molecular scale or allows the simulation of events at any level in biology on a sufficiently parallel scale to describe all of the things that are necessary to keep track of a particular system.

For example, bioinformatics has sprung up recently as a way to deal with, in part, all of the gene expression data that is out there. There are something like 6,000 genes in yeast and they turn on and turn off in response to certain conditions. First of all, how do you keep track of all that information in a way that makes any sense? Second, how do you represent that data in a way that you can interact with it? No one had to face that problem in biology until recently. It's still not the case that you necessarily have to keep track of all that information. No one knows, for example, how many genes you really have to keep track of to understand any particular given trait. One way to find out is to knock out a particular gene, thereby knocking out the protein, see what gets mucked up, and begin to make a list of what is important that way. While that strategy works, it is rather piecemeal. Often a gene we thought we understood is found to influence a trait or pathway that has nothing to do with what has been discovered previously. But that's the way biology has traditionally been done.

Now you can start to go at it from the other way. You can keep track of everything and figure out from the top down what part goes where, and when you wiggle a particular gene how many other things change. One answer to your question is that the measurement techniques just haven't been there, up to this point. It's also true of things on a more molecular scale. How do you carefully model the behavior of a single protein in a cell, or a single species of protein in a cell, when it's very difficult to monitor within that cell what that protein is doing? There are lots of techniques coming online that allow you to watch one protein as it does something within the cell, not just extract all the proteins from the cell and after the fact infer what happened. Now you can actually watch things happen.

New measurement tools, and the ones we still need

GBN: What exactly are these new tools? What are some of the new techniques you're referring to?

Carlson: Here's an example of a tool. You might attach a florescent molecule to a protein—effectively, making it a light bulb. If nothing else in the cell shines the way that light bulb does, then you can watch where that protein goes inside the cell by keeping track of where the light bulb is. We also now have inexpensive intense light sources to illuminate the light bulb and inexpensive sensitive detectors to measure where it is. That's a specific kind of technique that has not been around forever in molecular biology. It's relatively recent. So that provides information about what that particular protein is doing in a way that now you can go write down some things about the protein and maybe make a more physics-like model that you couldn't formulate before. It is true that cheap computers and cheap software and the ability to handle large amounts of data are making that task easier—at the genome level, the bioinformatics level—for many people.

But from my perspective as a physicist, it's really the measurement technologies that are driving where biology is going at the moment and how biology will develop in the long run. It's very easy to tell stories about something when you have no data; you can say whatever you like. In some sense, as a physicist, I'm a little more sensitive to that in biology than other people are because there's been an embarrassing history of physicists strolling into the biology lab and saying, "Oh, this problem is easy. I can solve that, no problem." A few years later they go away with their tail tucked between their legs and the biologists laugh.

GBN: Meaning that the physicists didn't have the data?

Carlson: Meaning that the physicists couldn't solve the problem for whatever reason. There has been a lot of ink spent and trees cut down to tell stories in theoretical biology, especially by physicists, when a lot of those stories were completely unconstrained by data. There was no way to test any of those models because you couldn't measure any of the parameters that were predicted.

Biologists have made progress, in contrast, by relying on data and relying on experiments and formulating what comes next based on the limits of the tools they had at hand, rather than the physics approach, which is to treat it more as a black box and say, "What's possible?" Now, because new measurement techniques are available, that means the physics approach is becoming ever more useful in biology. You can step back and say, "How does this thing work?" with the knowledge that you will be able to measure—today, perhaps tomorrow—the thing that you're talking about, and biologists are not used to thinking about problems in that way. That's definitely a distinction that I notice between physicists and biologists.

GBN: To simplify it, is it safe to say that biologists are following the front line of what's observable and measurable, but not necessarily taking the leaps into theory beyond what the data is saying, while the physicists tend to have theories which they then go test and find the data? But since both need the ability to do that measurement, they are coming closer to the same place.

Carlson: That's a fair way to say it. There are a lot of good theories in biology. A lot of good, general stories about how biology works have been told by biologists over the years. The reason they are probably more successful in telling stories about biology than physicists have been is that traditionally they have been more linked to the measurements that have been made and that can be made very soon. The reason theory works so well in physics is that there's this funny relationship between mathematics and the world that we see when we look out the window. For some reason that's still mysterious to many people, including myself, physics actually works. It's not at all clear why when you write down some equation that it has anything to do with what you see when you look out the window. It's a remarkable thing, actually.

GBN: The theories just don't include that life you see outside the window?

Carlson: Not including life. Well, including life, for that matter—it's just that we have no theory for life. The ability to write down those kinds of mathematical descriptions doesn't work at this point for the tree we see out the window. It works for the building over there, for all the stresses that have to be calculated to make sure the building doesn't fall down. You have to have some kind of quantitative predictive model that allows you to make a pretty good estimate of what the building will withstand so that it doesn't fall down when a magnitude 7 earthquake comes along. That same kind of framework does allow you to say something about the *mechanics* of the tree, what the tree has to do in order not to fall down, but tells you very little else. There are all kinds of stuff happening in the tree that are much more complex than what is happening in the building.

GBN: And we don't have a formal way to figure that out.

Carlson: We don't have the theoretical framework, and we're only just ("just" meaning 10 to 15 years from now) developing the measurement tools that will allow you to say a lot about the molecular details of what's going on inside the tree.

GBN: What kind of new tools are we talking about? Have there been a series of breakthroughs?

Carlson: Specifically, here at the Molecular Sciences Institute, we're working on a signal transduction pathway in yeast. That is probably the best understood eukaryotic pathway. Eukaryotic means cells that have a nucleus. As such, that pathway is an excellent model for human systems, but the yeast version is a great place to start because yeast are relatively easy to work with in the lab and the American Society for the Prevention of Cruelty to Animals doesn't get upset with you for any reason—thus far, anyway.

The models that have come before this time, for this particular system, are generally models that involve natural language stories. There exists a large amount of data; the model resides in the results of experiments. When we knocked out this gene, this thing happened, so we know that this protein is implicated in this particular kind of function inside the cell.

Here is the prerequisite for changing that story. I want to be able to know that when I add 100 molecules of alpha factor, which is the molecule that this particular signal transduction pathway measures in the environment of the yeast, that these three particular genes will be turned on and that 100 copies of protein X will be made, 300 copies of protein Y will be made, 400 copies of protein Z will be made, *and* that preexisting copies of protein P, which were involved in measuring the amount of alpha factor, will become chemically altered and will run off to the nucleus and change the transcription of other genes in a quantifiable and predictable way. Say you've measured the effect of adding 500 molecules of alpha factor and say you've measured the effect of adding 10,000 molecules of alpha factor. We want to build a model, a mathematical model, that describes what happens in detail there. You can then predict what happens when you add 5000 molecules or when you add two molecules. The prerequisite for the ability to develop that kind of predictive model is measuring on a more detailed scale what happens to protein P when you throw in X molecules of alpha factor. Because the tools now exist to attach a light bulb, a florescent probe, to protein P, you can watch it do what it's doing. There's a specific example.

Here's another example. It's now possible to make, in a relatively short time, artificial affinity reagents called aptamers, sort of like antibodies, that stick to proteins and that interfere with the function of those proteins in the cell. In the human body, you make antibodies against foreign proteins and that's one of the ways your immune system recognizes foreign molecules, and foreign cells, in the body. It would be useful inside a yeast cell, for example, to be able to tag different kinds of proteins and watch where they're going using a similar kind of system. If you had an antibody that you could stick inside a yeast and grab on to protein P and tell you where it was instead of having the permanently labeled protein P, that would be a useful kind of tool.

That exact kind of tool is being made at the Molecular Sciences Institute and other places—a sort of probe that allows you to isolate what is going on in a cell and watch. I'm focusing here on things that appeal to our visual sense of measurement. You can watch what's going on in a way that we haven't been able to before. But there are other tools that exist today that didn't exist even a short time ago that are more traditional biological tools, such as reverse transcriptase, which is an enzyme that some viruses use to turn RNA back into DNA. That's a tool that we didn't know about before studying HIV. It is now used everywhere in laboratories to turn RNA back into DNA so we can better measure how much RNA is in the cell, which is another important component of figuring out how information is shuttled around and how decisions get made inside a cell. It's just much easier to work with DNA than it is with RNA, so if you have the ability to work with DNA instead, then you're going to choose that. We didn't have that ability until 10 to 15 years ago.

GBN: When you're using the word "tools," you're not necessarily talking about physical microscopes. What you are really talking about are biological tools. These are rather like living tools—chemicals and such.

Carlson: To some extent, that's true. I'm also talking about electron microscopes. I'm also talking about atomic force microscopes. I am talking about all those things as well. We could talk all day about the different examples. The class of tools that is going to be most important in the future is going to be biological widgets, probably. There will also be some important fusion of, say, the atomic force microscope and an enzyme that measures I-don't-know-what, but in a way that no one has ever pulled off before. That sort of thing will happen, the fusion of biology and our ability to manipulate matter at very small scales that has developed from microelectronics and lithography. There will be a fusion in biology that leads to biosensors, or to labs on a chip, whatever you want to call it. More exciting and interesting for me, in the long run, is the use of biological molecules, which we learn about from studying biology, to then manipulate biology with greater facility.

Accomplishments of the last decade (human genome) and what's coming in the next decade (bioterrorism)

GBN: Is there a useful framework to think of how biology has evolved in the last couple of decades and where it might be going in the next 10 years?

Carlson: The measurement tools I talk about probably took somewhere between five and 10 years to get going. An example is the atomic force microscope: The first application wasn't to stick a cell under an atomic force microscope because the technology itself had to develop. The atomic force microscope had to be developed to the point where it could be used above liquid nitrogen temperatures. Initially, it could only do limited things—like look at frozen samples—whereas what you'd really like to do is watch enzymes move and that sort of thing. You can now do that with an atomic force microscope. But really, that sort of work probably started only five or six years ago, maybe a little bit before that.

Looking into the future, it's very difficult to predict the course of technology. One of the important lessons about sequencing the genome is that when the project was started there were some people who said, "This is absurd; it will never get finished. It's just too big and too hard." They were saying that in the context of what tools existed at that point. That was around 1990. There were other people who disagreed. They could conceive of a situation where, using the tools that existed—in particular the ABI 377, a kind of gene sequencing machine—they would buy a lot of those machines and hire a lot of people to sequence the genome. There was always the thought out there that you could spend a lot of money and solve this problem, using those tools. Some people were imagining \$3 billion. Others were imagining \$10 billion. Some people were imagining \$15 billion and 10 or 20 years to even make a dent in the problem.

Then along comes Craig Venter, who says, "Well, actually, technology is moving fast enough. There's a new machine that not only operates more quickly than this previous machine, but it can be automated in a way that was not possible before." They could remove human labor from a lot of the processes. Moreover, he realized he could use a different strategy for sequencing. So suddenly Craig Venter starts, and 18 months later he's finished. The human genome is effectively finished in 18 months. It's true that he used data from the government project to help provide landmarks for his efforts, but nonetheless, he finished that project for something like \$200 million, and it was finished long before even the original optimists said it could be done and for much less money—because of the way technology changed.

GBN: Is it safe to say that the '90s were about getting the genome mapped and this decade will be something else? I am curious about what you think could be coming in the next 10 years, with the caveat that no one can anticipate what might blindside the field.

Carlson: The point I was trying to make is a little bit different. Here's what I mean: The world is going to change a lot faster than we can imagine. It's not just that I don't want to look out into the future and say what will or won't happen. What I am trying to say is that things are going to be wild and crazy much sooner than we expect.

There are a couple of reasons why I say that. One is how fast our ability to manipulate biological systems is changing. If you look at our capability to either sequence DNA or synthesize it from scratch, then both are improving at least as fast as Moore's Law. Assuming those trends continue, within five years all of the sequencing effort that went into the human genome project will be accomplishable by one person on the laboratory bench in one day.

There's no reason to think that won't happen. That's just the way the trend is going. We're already really beginning to use biology to manipulate biology—we have for some time—but now the power to further manipulate biology, in terms of the biological widgets we use, is moving very quickly. What I see is that within just a few years we are going to have to contend, in a way that we don't right now, with organisms that have been modified in fairly sophisticated ways. This is particularly important when you consider bioterrorism; I think that's an important point.

I'm a physicist, so I have my own particular perspective on biology. In truth, what I really do best is build things, so biology for me is another set of widgets for building. And, as people like engineers get involved—there's now a growing group at MIT who are involved in the molecular engineering of cells in a hard-core engineering sense—we are going to learn a lot more about biology.

Carlos Bustamante likes to compare what's going to happen in biology to what happened in chemistry. Carlos Bustamante is a professor of physics and biology at UC Berkeley, and he does a lot of work looking at how single molecules behave, looking at the physics of single molecules. He's also interested in building things out of biology: building cells from scratch, building autonomous systems out of biological components from scratch. Carlos talks about synthetic biology in analogy to synthetic chemistry. In the early 1800s chemistry was more about understanding how things that were already out in the world were put together in a chemical sense: how chemical reactions happened, the structures of things, the chemical makeup of things. There weren't that many people who were interested in making new things. It was thought by many that organic molecules were beyond synthesis, that they were animated by a vital force. Then along came Friedrich Wohler, who synthesized urea from scratch. Suddenly, the way chemistry was done changed because you actually have to understand more details about the system to build a thing than to simply look at it and poke at it and describe how it works.

It was the process of moving to synthetic chemistry, and then an engineering of chemistry, that radically changed the way people understood chemistry because they had to learn lots of new rules that weren't apparent before. In the same way, as we move into engineering biological systems, we will learn a lot more about the way the pieces work together because challenges will arise that aren't obvious just from watching things happen and we will have to understand and address them.

What I see in the next few years—and this may be a good point to turn toward the terrorism issue—is that we currently spend a lot of time thinking about how to respond to things that are out there now: our approach to disease, our approach to smallpox. Much of the money that has been allocated to fight bioterrorism was spent on smallpox vaccines and buying Cipro to combat anthrax. What is technologically feasible today is much greater than what has been accomplished recently with anthrax. Before these attacks, the bioweapons experts were saying that it takes a rather large amount of anthrax to cause an infection, that the spores have to be very carefully manipulated. But we have examples where people seem to have been infected by a very small number of spores and died as a result. We have a transmission technique that no one really guessed at. We have a public health diagnosis system that was inadequate, to say the least. It was difficult for people to find out what they were looking at or what they were looking for. There was no way for people to talk to each other and find out what was going on. But our response is still to stockpile antibiotics and vaccines.

That's what I mean by we're solving yesterday's problem. We're not solving the problem today, which is that anthrax infects people much more easily than expected. It's much easier to distribute than was expected. We're not doing anything about that at the moment. People are potentially going to irradiate all mail, but this only addresses a specific problem. It doesn't solve the general problem that biological agents are out there in the world and can be distributed.

Only a bit further in the future, we should be concerned that within just a couple of years we will be seeing modified organisms that have much greater destructive potential. Within six months this lab that you're sitting in right now could probably make a bug that is incredibly destructive. And if a bug like that was released, we don't have the ability—we being scientists, we being biologists, we being the public health system—to do anything about it. We do not have the facility to find the bug, to isolate it, to understand its biology, to develop a response in time to do anything about it if it's sufficiently virulent. The skills exist today to make that bug. They don't exist to do anything about it.

GBN: Tell us, more systemically, why that is. Why is it a lot easier to make it than to understand it?

Carlson: I will step back and briefly explain why that is. In six months this lab could make a very nasty bug, which means that there are a number of labs around the world—not a great many, but a number—that could make a bug like that in six months. I'm not worried about any of the labs that currently have the capacity today to do that, but because we have it today, in two or three years everyone is going to have that capability if they want it.

GBN: What's on the verge of happening?

Carlson: Here's the horror story, a specific example. Let's say that a bug popped up today that grows like the organism that causes tuberculosis, *Mycoplasma tuberculosis*, which means that it takes at least two weeks to grow enough of the bug in the lab to do any biology with it. That's important because most off-the-shelf laboratory tools in molecular biology work only with large numbers of cells. Next, if you were going to completely twist an organism into making it do something bad, it's probably something like *E. coli*, but whether it's a *Mycoplasma* or *E. coli*, it's going to have some genes we understand because those are the genes that are required to live like every other organism on the planet. Now let's say it also has a bunch of genes we haven't seen before. Not all those need to be real genes. Some of them might be noise. We don't have a way to figure that out quickly. We don't have a way to figure out which of those proteins does whatever it does that makes the thing pathogenic. Let's even say it's not an artificial pathogen. Let's just say it's a pathogen that we've not seen before. Let's say it's something like a new Ebola or some other bug.

Now let's go back to the hot zone fantasy of a few years ago; a bug comes out of the jungle that we've never had to deal with before. What's our response? Quarantine. We can't do anything about the pathogen itself. It takes a while for us to go through the motions of sequencing the genome, and to do the rough biology to find out how the organism works in order to defeat it. We have to develop the ability in the next few years to go through that process much faster, to be able to look at how the pathogen interacts with its host, to look at what proteins are there and what they do differently that we've never seen before.

Now let's go back to thinking about an engineered bug. In this lab, let's say we used PCR shuffling or something like it to alter existing genes, or to make some new genes, and put them in a bug. It would probably take us six months to make a thing that was deadly if we set our minds to it. Because we can do it here, because we have the skills and the machines sitting around and the protocols written down and available to us, the raw materials on our bench, in two to three years the technology will be sufficiently distributed that many other labs will have that capability. When I say lab, it doesn't have to be a high-tech government, industry, or academic lab, but technology in two or three years will probably be so good that you could do it on a countertop in your kitchen. Molecular biology is high tech in some ways, when you look at the robots moving around, etc., but a lot of the fundamental things you do at the bench are relatively mundane. They do require some skill, but skills can be learned.

GBN: What you're implying here is that the best defense is an offense. You need to figure out the predictability and how and why things work so that when things inevitably surface, we can respond to it immediately.

Carlson: That's right. The measurement tools and the models that we will be using to understand and manipulate systems are exactly those tools we need to diagnose things when they go wrong and to do something about it. They are also the tools we need to make improvements in human health and to cure human disease. There is no question that those tools will be developed. It's not like we can say, "Whoa, we're not going to allow anyone to work on synthesizing DNA faster or we're not going to allow anyone to sequence faster or we're not going to allow anyone to understand protein networks faster." Those things are inevitably going to happen, and because they will be developed, and are also useful for manipulating systems, they will be disseminated.

GBN: Is there an end point where the ability to counter this stuff will get the upper hand? Will it always be a biological arms race?

Carlson: The ability to fix the problem will arise and we won't always be under the gun. We won't always be struggling to catch up. At some point there will be a time when it will be OK that people will screw up or that people tend to make mischief, because the technology will exist to solve those problems and solve them quickly. The reason I say that is not because I am an optimist but because we don't have any choice. We can learn how to solve problems quickly or we can not learn how to solve problems and suffer the consequences.

I fully believe we will have the capabilities to manipulate biological systems in great detail, and as part of that capability we will be able to respond to potential threats that arise either through mischief or through mistake. I could speculate all day long about the wonders I see down the road.

GBN: Do you see a danger zone in the next decade?

Carlson: The danger period is when we can manipulate biological systems but we can't do anything about those manipulations—when we can't go back and fix things. We are already in that period. It is today.

How long is that going to last? Well, we estimate that we'll be finished with the project here—with the yeast mating pheromone transaction pathway, the Alpha Project, as it's called—in about five years. That means we will have a model that accounts for the behavior of 10 to 20 proteins in a quantitative, predictive way. That is, we will be able to tweak the system in ways that we've not tweaked it before and be able to predict the response. There will be fruits of that effort all along the way that include new measurement tools, manipulation tools, modeling tools, and simulation engines. At the end of that, the collection of the results becomes a set of tools that are immediately applicable to just about every other biological system.

There are other researchers elsewhere in the world attempting to do the same thing, without necessarily having the exact same goals, so in roughly five years we are going to have a very diverse set of capabilities both in understanding and manipulating biological systems. There will be a period where we will enable more mischief than we will be able to do anything about. There is no question about it.

There are people who argue that we ought not to be doing this work. But as I said before, we can't stop it. Even if we could distinguish between those technologies that are critical to improving human health and those technologies that are only useful to make a weapon, which I think is a distinction that cannot be made, the United States cannot mandate how those technologies develop. They are sufficiently widespread at this point that development will occur whether or not we proscribe particular research in the United States.

One example of that is from Charlie Cantor, who is at Sequenom and on sabbatical from Boston University. He was at a meeting at the Chinese Academy of Sciences a few years ago, and the head of the Chinese Academy of Sciences said that as soon as they could, they would be using all means available to them, including genetic manipulation, to improve crops and the lives of the population. It's certainly true there are capabilities in the U.S. that don't exist in China, but it's also true that they're not going to listen to us if we say, "Don't do that." They will do it, because of their population base, and because they are making an effort to bootstrap themselves technologically. They will also be developing things simply because they will have more people working on more projects. They will be developing things that we're not working on here necessarily. So new things will happen in China that we don't have any say about. That's just one example.

Slowing the aging process; intellectual property and the problems of patents

GBN: You make a compelling case for full-speed-ahead here. What else is worth preparing for in this decade?

Carlson: I can probably come up with a very long list of things. It's important to remember that my perspective is different than most biologists. If you get molecular biologists drunk, or at least get a few beers into them, it's not rare for them to say they'll have aging solved in 20 years. That's not something they say in public. It's not something they often say in print or even speculate about in print because aging is extraordinarily complex; development is extraordinarily complex. But when they scratch their chins and let their guard down a bit, they speculate that we'll have a sufficient understanding of those complex systems within 20 years to be able to make significant changes in how the systems operate.

It's also true that people tend to say it will take two to five decades in order to solve the big problem they're working on because that timeframe is beyond when they'll be working on it. It then will become someone else's problem. That certainly is true in physics. If you ask an older physicist when a problem will be solved they often will tell you, "Well, about 50 years." They give you a timeline that is within your career but past theirs—to give you some hope, maybe.

GBN: They think they'll solve it in their lifetime or pass along the problem to the next generation?

Carlson: There are both examples. One is they put it out far enough so that they feel a little more comfortable speculating about it. It doesn't have to rely so much on the details, how hard they may think the problem is right now. A lot can happen in 20 years. Or they'll say, "Well, young man, in 50 years it will be solved." It's sort of saying, "You'll have it solved in 50 years."

Anyway, there are important issues to keep track of and there are things that are just going to happen. One of the important issues is intellectual property. The decisions we make now could strongly affect the availability of technology down the road, the availability of particular genes that are patented by one company to be explored by another company that doesn't have the patent. Currently, if you're trying to understand how a particular protein works and some other company has the patent on it, you can't do anything that has any economic consequence without getting a license first; otherwise you are infringing on a patent. That's a problem.

Personally, I think the government is making a severe mistake in granting patents on natural genes. If you're talking about a manipulated gene, something you've added value to by putting some effort into altering it in some way—altering a pathway, altering an organism—fine. I don't have any difficulties with patenting that. But patenting a gene that preexists in the environment is rather like patenting the fact that trees are green. One argument for granting patents on natural genes is that the cloning of a gene, the process of copying it from the original organism to a more portable and useful form, necessarily involves an "improvement," an alteration. But what would have happened if J.J. Thompson had patented the electron, or if Einstein had patented the photon? J.J. Thompson certainly did something novel with electrons, putting them literally in a new state.

The concept of gene patents also doesn't make sense to me from the perspective of innovation. What happens if there is a Biosoft, a bio version of Microsoft that develops and sells molecules and technology that constitute a biological operating system? I guess there are really two issues here. The first is innovation. What happens if one company has control of an entire system of genes that is key to, say, human reproduction, or even bovine reproduction? Or wheat?

The second issue is the quality of the technology and the effects of proprietary technology. What if there is a mistake made and not enough other people have sufficient experience with the system to quickly respond? There is an old joke about Microsoft and embedded systems, that is, the notion that soon computers will be in your washing machine and your toaster and in every system in your car. Those computers will need some sort of operating system. The joke is this: given your experience with Windows, do you really want Microsoft in your car's brakes? The analogy to biology is clear: do you want Biosoft in your food or in the protocols and molecules we use to fix genetic diseases? This isn't radio, or television, or the look and feel of an operating system. We are talking about the stuff of life. Again, I don't feel it's a realistic option to try to prevent research and genetic modification, but it is well worth thinking about the larger environment in which such work takes place.

This is something that I am trying to spend more time on. I've written a bit about it, trying out the idea of "open source biology," in analogy to open source software. There are enough people worried about it that something positive will probably get done, but it's not something to take for granted. The way we're going right now, everything is going to be locked up and no one will be able to do anything useful.

GBN: You are sufficiently encouraged that there's enough discomfort in the scientific community that a counterbalance could be applied and the issue could get solved in a more open manner?

Carlson: Yes. I think that the debate going on in the software community right now is an intriguing example of this kind of thing. I'm very interested in what happens with the open-source software community. As far as I know, to this day neither the General Public License, the GPL, nor the Lesser License, have been tested in court, which is interesting. How well will this structure hold up in the long run? This is important because I see much more innovation happening when information is available, that is, distributed, than when information is not available. There is also the example of Goldcorp mining, which opened its geological survey data to the public as part of a competition to guess where they should next look for gold. Traditionally that data is kept tightly guarded, but by distributing their data to the community, Goldcorp has made an extraordinary amount of money because they effectively brought in a wide range of experts and approaches. It is an interesting model for innovation, and one that might well work in biology. So, yes, I am concerned about loss of biological information behind closed doors, but there are apparently successful examples of other ways to go about it. Part of the answer to your question is also that I'm just stubborn and I'm not going to be satisfied with no answer. I'm going to keep doing something and have faith that I will have some impact even if everything looks truly dark.

There are people out there who are taking a bit of a different tack. There's a piece in a recent *Nature Biotechnology* about patent pools in agricultural biotechnology. It's a different way to handle patented genes, or, rather, to handle the fact that there are gene patents on natural genes. That's an approach from the corporate side. I'm thinking about it more from the innovation side rather than making money off a thing that already exists. How do you make a new thing? How do you use a thing that's out there already to make a new thing?

Stem-cell debate and political debates

GBN: At some level it's still an open game that everyone is trying to figure out. Will it lock into some kind of convention that's going to set the tone for the future?

Carlson: We can probably still decide what that tone is. Once that's set, I don't know how easy it will be to change, because suddenly a lot of money will be invested in the way things start to operate. That makes change difficult. Another thing I think will happen is that people will get stem cells working. It's a hard problem, but this time last year there was an editorial in *Nature* pointing out that 12 months previous to that, the developments of the year in stem cells had been science fiction. Within a year the scientific landscape was tremendously different as far as human stem cells go. A lot was known about how to make genetic manipulations of mice with stem cells, but none of that was possible with humans because mice aren't always the best models for humans, as it turns out.

Basically, every month something new comes out. Recently, there was a disclosure from Advanced Cell Technology that they had "cloned human embryos"—which maybe they did and maybe they didn't. We can't really sit down and look at the science because they haven't published it.

I think it's very interesting that George Bush's response was, "It's bad public policy and it's morally wrong." For me, that doesn't make a lot of sense; it can be bad public policy *because* it's morally wrong or it can be bad public policy *because* of something else, but he seems to have made his decision already about what public policy is when in fact there is no public policy yet. In fact, he seems to have conflated his opinion with public policy. Much more importantly, the basis for his decision is limited, as far as science goes, because we know so little.

Amidst the recent debates about stem cells and cloning embryos you often hear people arguing about when life gets started, what life is. Only once or twice have I heard anyone point out the fact that life probably only started once on this planet, at least the life that led to humans. I don't mean Creation, but rather biology. Then again, perhaps the argument works either way. The egg is alive; the sperm is alive. It's not like life, per se, begins afresh at conception—it's a continuation. It's true that the survival conditions in the womb determine when an embryo develops at what stage, whether it survives or not. What I'm getting at with Bush is that a lot of decisions are being made and a lot of public debate is happening right now without much reference to real science, to things that we can measure.

The hubbub is basically the same for genetically modified foods. A "foreign" protein that's put in a tomato is not made of anything different than what is in the rest of the tomato. It's just that it's put together in a different way. It does something different, which is not to say that we shouldn't be concerned about the effects of that new thing, but that should be the debate, the actual biological effects. It shouldn't be that a foreign protein put into a tomato is an affront against man, God, and nature, because that debate doesn't help us decide anything about what is really happening to us and the world around us. If that's the level we're going to debate things on, then OK, I guess we have to be satisfied with that, but we have a tradition, especially in the United States, of relying on science to some extent when we make decisions, even if it's not a perfect track record. At this point, most of the debates about using biology as technology revolve around hearsay and things so-called experts say - not necessarily with the aim of education but rather with the aim of notoriety. I think a lot of that will blow over as science goes forward.

GBN: You're saying that despite public policy, despite the pandering in the public debate, there's a certain default way it's going to go?

Carlson: Science will go forward; technology will go forward. The example everyone always gives concerns test tube babies. There were people back then arguing that test tube babies were going to be nonhuman somehow. To suggest that today is ludicrous. Even the people who were most adamantly against in vitro fertilization at the time can't stand up today in public and say, "Well, actually these people aren't human; they don't have souls," because that's clearly stupid. There was an honest debate at the time, even though the debate was uninformed by science. Making statements like that today is stupid because there's lots of science and everybody knows what it is.

What I am trying to say is that I expect the same thing to happen with cloning. That is, there's a kind of stupid debate happening today, which doesn't have a lot to do with science because not a lot of science has been done yet. Despite what people say about stem cells and cloning, the work is going to go forward—if not in this country, then somewhere else.

One significant feature of our public discussion is that most of our trusted publications, everything from *The New York Times* to *The Economist* to *Scientific American* to *The Wall Street Journal*, and most commentators to boot, tend to base their explanations, suppositions, and opinions about the future not just on *what* science has been published, but more importantly on the *way* science has been done.

There are two issues here. The first is a feature of scientific publishing: any result in the press is at least six months old, more likely between one and two years old. It takes time to write a paper, get it accepted in a journal, and then get it published. Science in the lab is often much further along than the current press would indicate.

Second, the technological infrastructure we rely on today is bound to change. The *way* we do science will change and the *way* we interact with and manipulate biological systems will change. For example, one argument against any sort of cloning that involves human eggs is that the success rate is very low. It takes many eggs, hundreds to thousands, to create one viable blastocyst using current techniques. But however you feel about using human eggs in cloning, the number required for success is bound to fall as we learn more, and as we learn to do things differently. Consequently, the debate we have in the future will be much more informed. It would be nice to leapfrog a little bit and have a more informed debate today. I am trying to work out how to do that myself, how I can contribute to that informed debate. We may just have to suffer through this period.

GBN: Give us your perspective on the potential in stem cell research.

Carlson: I will approach that by contrasting two different ways to grow replacement tissues.

There are today two kinds of technologies that might be used in growing organs. One way is to start with a small tissue sample of an organ you want to replace and grow it into something that is functional and useful as a transplant, and that work is already going forward without stem cells. Researchers are using donor tissues or tissues that are less subject to rejection. Work is already going forward on tissues that don't have a lot of structure. Bladders have been grown from scratch. Livers are being grown from scratch. Kidneys are under way. I know the lab-grown bladders have already been transplanted into animals. There's a project under way to grow a human heart from scratch in 10 years. Maybe now it's eight years, as the project's been under way for a couple of years. The human heart is complex and not all the project participants think that they're going to get it done, but nonetheless, that's the idea.

The other way to go is to use stem cells. There are a number of scientific issues involved. One is being able to isolate sufficient stem cells from any source to do anything interesting. The other is getting them to divide and develop to make the tissue you want, which is different from growing the organ from scratch. If you are growing a liver from scratch, maybe what you do is take a sample of a liver and figure out how to make those cells divide and grow them in an environment so that they more or less look like a liver when the growth gets finished. That's different than encouraging a stem cell to become a liver cell and then eventually a liver. Once you get it to become a liver cell, then it looks more like the previous problem of just growing a liver from a liver cell.

GBN: You're saying that the growing of an amalgam of liver cells that coheres and would function as a liver is already happening?

Carlson: That project is under way, and I believe that a liver has already been grown in a dish. People that work on this have already grown bladders and implanted those in animals, and they work.

There's a lot of tissue engineering involved—that is, do you grow it on a framework? What medium do you grow it in? What temperature do you use? The general approach in organ growth is to use a bioreactor. You start off with a small bundle of cells suspended in solution and you stir it in a way that looks more like the womb than it does the surface of a plate, for example, which can strongly affect the morphology of the tissue that develops in the end. There are researchers growing skin, for example, which is a complex organ. It's not as complex as the heart, but nonetheless it has different layers and different cell types, etc. The skin is grown on artificial frameworks that dissolve, and there are several companies that sell transplantable skin now that effectively has been grown up from a few donated foreskins into several football fields' worth of transplantable skin.

GBN: Wow. That's in the here and now?

Carlson: Yes, that's Advanced Tissue Sciences and Organogenesis. If I recall correctly, Advanced Tissue Sciences is growing arterial grafts from scratch. The idea is that instead of using plastic stents, you'll be able to grow a piece of artery in a dish and then stick it into a body relatively soon.

GBN: What is the stem cell's benefit over that?

Carlson: That the original cell comes from your own body. There's no question that it's biologically compatible with you in every way. But who knows? Maybe there's going to be some crazy problem where you take a stem cell out of your body, grow it into a heart and your body doesn't recognize that as you anymore. That's speculation, but then it's speculation to say that there will be no problems whatsoever along those lines. Development is complex. Who knows how long it's going to take to really get appropriate isolated stem cells? Who knows how long it's going to take to really cleanly get the development process to go forward in a way that makes sense and is useful?

Help for genetic disorders and other diseases

GBN: This is a pretty extraordinary time to be a biologist. You've just cracked the human genome; there have been these wild, heady advances in stem cells that surprised everybody. Then there's computer modeling at the level where you are starting to get systemic tracking.

Carlson: It's amazing. We could conceivably begin to manipulate ourselves in ways that are completely unprecedented. But when it comes right down to it, we still don't have a great understanding of a lot of diseases. We don't understand cancer, even though there are very promising approaches to getting rid of, for example, solid tumors using the immune system. There are very promising ways to go after pathogens in the body.

At the same time, most genetic disorders probably involve more than one gene. We only have a very few well-understood genetic disorders to begin with, and they all involve only one gene. So we have a long way to go. How are you going to start fixing genetic disorders that involve more than one gene? If you have two proteins involved, maybe you only have to worry about one interaction—that is, the interaction between those two proteins. But it might be an interaction between the first protein and 10 other proteins, the second protein, and 20 other proteins. It's not a trivial problem. This whole proteomics business is going to take a long time, at least with the tools we have now. Of course, other tools will come along.

GBN: There are a lot more proteins than genes—won't this project be a lot messier?

Carlson: That's probably true, but it's not clear. There are something like 30,000 human genes. The exact number is not yet pinned down. It seems like, when you count proteins using other methods, there are many more proteins than there are genes. Let's say there were just 30,000 proteins. It's not the case that you only have 30,000 things to go catalog now and understand. What you actually have to understand is the interaction of each protein with every other protein that it talks to throughout the life cycle of the organism. That's the kind of understanding that we're reaching for now.

Let's go back to the example of sickle cell anemia. Here's a single protein, hemoglobin. The difference between the diseased form of the protein and the normal form of the protein is a single amino acid, caused by a single change in the genome. Changing one letter of the genetic code changes one amino acid and that completely changes the structure of the protein. If that took, say, 40 years to understand and roughly 50 persons per year working on it, that's something like 2,000 person-years of effort to understand this one protein. That's not at the quantitatively predictive level that I was talking about before. We currently can't go from changing the base to understanding the change in the shape of the protein in a predictive way. Furthermore, we can't predict how the behavior of the protein changes in response. We can measure it all and describe it, but there's no model that allows you to understand how it all works. If we had to do that for even 30,000 proteins, we'd be at it for a very long time.

It is true, however, that technology is cumulative, and all the techniques that were developed to work on hemoglobin and every other protein that was worked on during that time period are available on the bench today. If you have a protein you've never done anything with before, instead of 2,000 person-years of effort from scratch, it probably takes two person-years at this point. If you're really good, it maybe takes a year. The story that results is not exactly the same kind of story that you tell for hemoglobin, which involves a lot of thermodynamics, a lot of statistical mechanics, a lot of structure. There's a huge amount of work that went into hemoglobin that you're not going to have to do for every protein that you want or need to understand in humans.

Nonetheless, it's still going to take a long time if we go at the rate we're going. We're going faster than we used to. We're likely to go even faster in the future, but what we have to do is not just tell the story of hemoglobin for 30,000 other proteins but actually understand how those 30,000 interact. It's not listing 30,000 things. It's listing the interaction of one thing with maybe 100 others and the next thing with 100 others—if that's as many interactions as each protein has. Maybe the average number of interactions is only 5. We don't know yet what that number is. The tools to do that quickly and in parallel don't really exist yet. But there are ways to go about it now. People use whatever they can to solve that problem, but that's definitely an arena for improvement. We could be going faster and doing better than we are.

GBN: Will this decade be about the protein project in the same way that the last decade was about the genome project?

Carlson: Now that the genome is sequenced, it's certainly true that one of the major tasks left is understanding what all those genes do and how they work. Whether that will take 10 or 20 or 30 years, I don't know. If we go at our present rate, then we are talking about hundreds of years. But it's also true that we are liable to go faster than we are going today. I'm not going to speculate on how much faster we might be able to go, because the next technology that comes down the road could radically change the way we do things.

People talk about how they like gene chips. Gene chips are sections of DNA on a chip, in a grid, that allow you to take a sample and spill that onto the chip and see if DNA from your sample binds to the opposite sequence on the chip. It's a way to figure out what is in your sample. The true utility of gene chips isn't really clear yet, but it seems that they will become a useful tool at some point anyway. Their utility, or rather the degree to which people get excited about them, has engendered lots of talk about protein chips where, instead of DNA arrayed in a grid, with different DNA in different squares, you'll have a different antibody in each square. What you will do is take your sample, spill it onto the chip and look for binding of proteins that were in the sample to antibodies that are on the chip as a way to assay for the proteins that were in the sample.

However, there are some big technical difficulties with making protein chips. One is that you have to have antibodies, or some other sort of affinity reagent, to grab onto the proteins. Each antibody has to grab only one protein out of that mix, not two; otherwise it's not a useful measurement. Then you have to be able to measure the interaction. You have to be able to see in some way that proteins from the sample have bound to the surface. That is not trivial, either. While chips are a potential way to do proteomics it is not obvious that will happen usefully any time soon because there are significant challenges. Then again, according to this week's *Science*, we might already have a functional protein chip. It's hard to say.

GBN: There are aspects that are out of control in all science, in biology in particular. There are so many people now with advanced degrees working in sophisticated labs all over the world. It's out of control, not necessarily in a bad, dangerous way, but in the way that no one knows what's going on in half of these places. It's pushing the boundaries on all of these frontiers.

Carlson: There is no constant but change. Here's another thing that will be very interesting to watch over the next few years. It wouldn't surprise me if a number of the things we call diseases—things that are just kind of mysterious now, ailments we say are just part of aging—are actually caused by pathogens, meaning a virus or a bacteria or something. We recently have added to the list of things that I would call pathogens, prions, for example, which seem to have something to do with Alzheimer's, or seem to have something to do with mad cow disease. Before 1980, no one had a clue that a protein molecule could cause an infectious disease, but it can. That's kind of an extreme example in some sense, and it's an obvious big red flag hanging out there.

In a slightly more subtle discussion—and it is a sufficiently sophisticated discussion that it has reached the pages of *Science*, for example—consider for a moment a disease like MS, which is not really a single disease but a syndrome, a collection of symptoms. When you display enough of those symptoms, an MD says you have MS. In a rather strikingly large number of cases of MS, particular viruses also appear in patients' blood samples. There's no cause and effect elucidated yet, but there is a remarkable correlation. It's not a pathogen that anybody thought a lot about 10 years ago.

GBN: What you're saying is that some of the diseases that we thought were more intractable could be viruses or combinations thereof that can be solved in the way we solve problems with other viruses.

Carlson: Actually, that's one of the problems. We can't solve viral issues very well right now. If it were bacterial, there would be another way to go about it. But I think what you said is very telling, in that we know about lots of viruses and there is very little we can do about most viruses at this point. If you look at the amount of investment in HIV, it's rather remarkable. We can do something, we can contain it, but we can't cure it. The treatments for HIV don't involve eliminating it from the body. They involve reducing its reproduction to effectively zero, if not actually zero. As soon as you stop taking treatments for HIV, you've got the full-blown infection again.

GBN: Is what we've accomplished with HIV astounding or disappointing?

Carlson: It's remarkable, but it's also frustrating that we haven't gotten any further with it. HIV is not an especially complex organism, compared with yeast, say, but it has held many surprises for us. We have learned enormous amounts about HIV in the last 20 years.

I will use another example to illustrate the point I tried to make earlier. Before 1990, ulcers were just ulcers. They were a problem of bad diet, or perhaps the conclusion was that you were genetically susceptible because your father had ulcers. Of course, we now know that approximately one-third of ulcers are caused by a bacterium and are treatable with antibiotics. Here's a bug causing a disease that for hundreds if not thousands of years was just thought to be some trait of being human and growing older. Similarly, we now have sufficiently sensitive measurement techniques to discover that there are viruses living in particular tissues in the body that we didn't know about five or 10 years ago, and the prevalence of those viruses is remarkably correlated with things that were thought to be just failures of the human system beforehand, such as MS. Now, again, I'm not saying that these are what cause MS—nobody knows that yet—but there's something funny going on.

GBN: Is it conceivable that within your career trajectory, disease could be understood predictably? How long until we can define diseases as to whether they are viruses or bacteria or a genetic defect?

Carlson: Let me phrase it a little differently. Sometimes around here we wonder if there is something like an Apollo program or Manhattan Project that has a 10- or 20-year time period, that would be sufficiently shocking that it would get people's attention but would also be sufficiently doable so as not to be ludicrous. One of the things that comes up in those discussions is eliminating pathogens as a factor in human experience. That is not a trivial undertaking by any means, but the goal is worthwhile. Or at least it is from my perspective. Of course there are people who would argue that we ought not to be mucking around at all with that sort of thing and that we should just let humans die as they will. Anyway, the task is both sufficiently rooted in detail and sufficiently devoid of detail that I can see that time frame being the right one for that goal. What I mean is, we already have the ability to measure lots of things, to do something about many problems. This gives me hope that, especially at the rate we're developing new solutions, within 10 or 20 years we will have a great capacity to influence infectious disease even though we now can't list in detail all of the problems we have to solve.

GBN: When you say pathogens, that would include all viruses, bacteria and toxins.

Carlson: Bugs, prions, whatever, yes.

GBN: But that wouldn't necessarily address fundamental aging processes, which is another ambitious thing?

Carlson: That's true.

GBN: A third possible category is genetic problems that don't depend on pathogens in the environment. Would you say those are three areas (pathogens, aging, genetic disease) that would be worthy of the moon project?

Carlson: I'm not sure that attacking infectious human disease has a specific set of technical goals and technical details that I feel comfortable thinking about. It's not like we can say, "In two years we are going to strap a guy in a tin can, shoot him 50 miles up, and hope he comes down alive." I think with disease we have to step up to one problem at a time. That is, there is a bug of some kind; it has a genome; it has some proteins; it interacts probably not with a large number of human proteins, but with a sufficiently small set so that it's optimized to work well. That's not going to be true in all cases. There are probably some pathogens that have a very diverse means of infection. But it seems like the problems can be broken down into solvable units in that way. Here's a pathogen. Here's how it works. OK, now we've learned a lot about that pathogen. We can solve that problem. When we step up to the next one, not only is it a smaller chunk than the whole problem, but we learned a lot from the last pathogen that we can apply to the new one. Of course, we can do this in parallel.

However, when you talk about human aging, I am not at all comfortable talking about that as an Apollo-style project because human aging potentially involves all of our genes and all of the protein interactions that make us up. When a molecular biologist talks about solving human aging in 20 years, they're sort of ruminating about how things have changed. Actually, it's been a remarkable 50 years. Fifty years ago today, we did not know about DNA, really. We knew that there were some interesting molecules and that traits were somehow listed in the DNA. We didn't know the code 50 years ago. The paper that Crick and Watson published only had the structure in it, but there's this line in the paper stating that the quasi-crystalline structure of DNA did not escape their notice as being useful for coding information. That's important. They could only *suggest* that it had some function, and that was merely 50 years ago.

When a molecular biologist says it will take 20 years to solve human aging, they're looking back and saying, "Wow, 50 years ago we didn't know that DNA is the carrier of genetic information." Twenty years ago we were just beginning to learn about AIDS and HIV. We have learned a tremendous amount of biology from studying HIV. Eighty years ago we barely understood viruses and bacteria and infection on a very descriptive basic level. Penicillin only happened in World War II, at least to a level where it became useful. But if you're really talking about solving human aging, it's not going to be simple. Even with remarkable results in yeast, fruit flies, and mice, it won't be trivial to transfer such results to humans.

Some people say if we can make fruit flies live 10 times as long, that means that we can make a human live 10 times as long. It's important to note that the scientists who do these experiments generally don't even claim that a fruit fly can live 10 times as long, but clearly, once the experiment is successful, someone else will stand up and say that all they have to do is live on a starvation diet and take a pill that changes their X enzyme and they will live to be 1,000 instead of 100. But it's so complicated.

What I can envision is saying, "For 20 years we are going to throw several thousand people at the problem." At the end of 20 years, they might have some things done that could strongly influence human aging. But to set that as the goal is overreaching, from my perspective. The reason I say that's the way to solve the problem is because that is the way we'll do it anyway. We are going to throw however many thousand molecular biologists at problems that are associated with human aging.

GBN: The three areas to consider are the aging bucket, the genetic manipulation bucket and the infection bucket. What do you make of the genetic bucket?

Carlson: It's like the aging bucket in that you're talking about a system that potentially has several hundred thousand moving parts and who knows how many different ways those parts interact with each other. Understanding that system is going to be a serious undertaking, but that's what we're doing. Now I cannot get my brain around solving the genetic bucket, but going after little tiny drops in that bucket is reasonable. Cystic fibrosis, for example, is a disease that results not from a single mutation, but from many mutations causing different molecular effects with the same basic resultant physiology. It's not just one thing that causes cystic fibrosis.

GBN: One thing, or one gene?

Carlson: It is one gene, but it can go wrong in a number of ways. It's not the case that it's a single nucleotide polymorphism, as it is for hemoglobin in sickle cell anemia. That is, it's not one base that gets changed; one of many things gets changed. Over a thousand unique mutations have been described. So one solution might be to go in and replace all of that DNA with genes that aren't "broken." That's gene therapy, and in the way I just phrased it, to replace that section of the genome in every cell in the body would be a significant undertaking. That would be human germ line engineering, which is a big deal—both technologically and ethically—but I think we will be doing that in 10 years. The fuss will have sufficiently blown over. When you talk about being able to eliminate human suffering through genetic modification, people are going to make much less noise than they have about corn or strawberries that don't freeze because those berries have a fish gene in them that prevents nucleation of ice and that sort of thing. So, there are bits and pieces of the genetic bucket that I can easily see we'll go after specifically, and that we'll solve.

If you talk about curing cystic fibrosis in an adult—if the patient manages to live that long—you're probably talking about genetic modification of the cells in the lungs in order to change the proteins that are produced in the lungs, which effectively changes the function of the living person. That's already happening. People are already working on various DNA delivery techniques for gene therapy, for alteration of the cells that already exist in your body. There is some really interesting work involving delivery of naked DNA that is taken up by cells in salivary ducts. This would be useful for maladies that might be treated by non-localized gene therapy. Basically what you do is infect some cells that live for a short time but aren't around that long, and they make what you need for some time and then they die and the modification goes away.

GBN: I was expecting you to say that kind of gene therapy for adults is not possible. Is it possible?

Carlson: It works in rats and mice. Human trials are beginning.

Genetic engineering

GBN: It would seem that in a fully-developed organism, it would be harder to affect the gene composition. Is it easier to start with a newborn or an embryo, at the beginning of the line?

Carlson: It depends. We don't know how to do germ line engineering in humans with an egg yet. There are people who are certainly playing with that, for example inserting a whole artificial chromosome in a mammalian egg. That's a way to include not just one gene, but a whole set of genes. It means making a human with not 23 pairs, but 24 pairs of chromosomes. It's not a brand new idea.

GBN: What would happen with 24 chromosomes?

Carlson: This work is far from finished, but it's already the case that you can put an artificial chromosome in a mouse—give a mouse an extra chromosome and the mouse carries the chromosome around and it divides normally and does everything like a chromosome. You can't make a chromosome that looks just like the other mouse chromosomes yet. It's not exactly the same structure, but it behaves in the same way.

GBN: Does it exhibit itself in the organism somehow, even though it's outside of the mouse gene?

Carlson: During cell division, which is an extraordinary ballet of components inside the cell, the chromosomes line up along the center of the cell and they're copied and pulled apart across the sides of the cell. It's just extraordinary that it happens, first of all, and extraordinary that we can watch it happen. But if you add another chromosome to a mouse cell, then you can watch the new chromosome do the exact same things as the natural chromosomes.

GBN: And the organism that grows out of that replication expresses whatever extra trait, or could.

Carlson: I don't know the extent to which a new gene has been carried and effectively expressed on that chromosome. I believe it has.

GBN: Is that how the bunny in Switzerland gets the fish gene?

Carlson: I believe that was a more traditional style of genetic manipulation. They took a whole bunch of either eggs or embryos and then randomly inserted the gene for GFP [green fluorescent protein] into the genome. They first selected the ones that survived, because sometimes that GFP gene was inserted into a critical gene elsewhere and halted the development of the embryo. Then they further selected for the embryo that expressed it in a way that would work. Who knows how many bunnies that grew up had the gene but didn't actually glow green and didn't do anything interesting. That's the same way that ANDi the monkey was made at the Oregon Primate Center. A few years ago the first primate was created that had an artificial gene in it and it was the same GFP gene. The researchers basically inserted it into a bunch of embryos and waited for the one where the placement was useful, which is a very different strategy than using artificial chromosomes, which is much more like genetic engineering. The phrase “genetic engineering,” the way it’s done now, doesn’t make any sense. There’s really no engineering about it. In some circumstances you can stick a sequence specifically where you want it and have the effect you want as a result of that insertion, but there are very few.

GBN: It’s more like cooking.

Carlson: Even worse. I had a letter in *The New York Times* about this a while ago, where I said that it was more like swapping parts randomly between cars and hoping you get a better car.

GBN: Which is a long shot.

Carlson: It’s not that you take one egg and genetically engineer it. That’s not the way it works. You roll the dice and hope that one of the things that you’ve changed in a random way is good enough that it does what you want. I should say, by the way, that this is changing. People clearly recognize it’s not the way to do things if you want to actually have some understanding and control of what’s going on. There is a lot of work ongoing on ways to insert sequences where you want them in every organism. Previously, it was possible in a few particular circumstances, and within the last year there have been some interesting developments. It looks much more likely that you will be able stick a sequence where you want it in the genome.

All the stuff we don’t know about and have yet to measure

GBN: Which brings us full circle back to the predictability. The biosphere is so amazingly complex and well designed that the idea that just random bumping around would create it seems ludicrous. I read an analysis that claimed when you play out the random evolutionary possibilities that it would take to get to the level of sophistication of our biosphere—well, it’s impossible. It couldn’t happen in 14 billion years. It brings you back to design, to the space of science and spirituality. As a young scientist faced with that beautiful dance in the cells, how would you respond?

Carlson: There are a couple of ways to respond. A simple retort is that it’s not impossible because here we are. But of course this begs the question of how we got here. A slightly more sophisticated retort offers a specific example. There is a recent book that claims that, given what we know about the universe at large, we basically have to be alone. The authors—cosmologists—argue that based on even the limited experience we have of looking out and measuring that humans have to be alone in the galaxy. They suggest that there is probably no other intelligent life around and there probably isn’t any other life at all in the galaxy. This hypothesis is based on astronomy and some theory. These seem like very strong statements to me, given our current measurement abilities.

It is certainly true that making hypotheses is part of science, and it is part of the fun of science. But we can actually say very little about the way things are out in the universe, the details of how various materials are distributed or the possible processes by which they are formed. All we can do is try to create consistent descriptions of nature where we find it. Our notions about what happens elsewhere in the universe are based on our limited experience here on Earth. If we have yet to imagine a given physical process, then by definition we can't include it in guesses about what is happening elsewhere. But time and time again we have discovered new physics and chemistry here at home only after trying to explain something funny we saw through a telescope. What I am trying to say is simply that I think we don't yet know very much.

If you really want to dig deeply into it, then everything we say about the universe comes back to the Cosmological Principle, also called Mach's Principle, which says that physics is the same everywhere and always has been. If that's not true, if the physics that describes what happens here on Earth is not the same as what occurs one light-year away or 100 light-years away or across the universe, then we can't say a damn thing about what we see in the telescope. If the physics that we see today is not the same as it was a billion years ago on this planet, then even the things we say about this planet are tentative at best. If the Cosmological Principle doesn't hold, then we're kind of screwed. We can't say anything about anywhere, including where we are now. From what we can tell, physics *is* the same everywhere—and that's of course the basis, to some extent, for the argument I mentioned that we must be alone—but I think there is still a very long list of things that we have yet to measure.

Every week in *The New York Times* or *Science* or *Nature* there is a new story where astronomers or astrophysicists announce that they've just seen this or that for the first time, forcing them to reevaluate their fundamental understanding of this other thing. While I firmly believe that physics has a lot of power, any power that it has fundamentally resides in our ability to measure things. If we can't measure anything, or if we have yet to measure something, then by definition we can't make conclusions within the framework of a quantitatively predictive theory and expect to have a result that we can compare to anything in the world. The ability to measure is required to make a reasonable prediction about a thing, which is why you often get comments that string theory is so far just philosophy. But, here again, science is changing and where for many years string theory didn't produce any measurable predictions, both theory and experimental technique have now progressed to the point where some hypotheses can be tested.

GBN: Interesting. I was talking to my colleague Peter Schwartz about the new evidence that the universe seems to be expanding at an accelerating rate rather than slowing down, which means that there is some kind of dark energy out there that is pulling it or pushing it. We can't figure out where it is. Reflecting on that, he says that the thing about dark matter, which you can't find but must be there because of our theories, is that because you can't measure it, we don't know what to do with it. A hundred years ago we didn't have radios, but that didn't mean that radio waves weren't there.

Carlson: That's exactly right.

GBN: Then we get radio and find that we're surrounded by radio waves. In a sense we are waiting for similar breakthroughs that will make sense of our outstanding questions about the world.

Carlson: If you look from the physicist's perspective, if you look at the kinds of technologies we use today, at the origins of our ability to use them as technologies in physics, then you can say something interesting about electromagnetism and the emergence of quantum mechanics. Our ability to make the transition from early applications like radio to solid state physics, etc., occurred when quantum mechanics reared its head as a theory to explain experimental results and measurements that were confusing, to say the least.

GBN: So quantum mechanics was about our making sense of radio.

Carlson: There was a conflict between the way people thought the world worked theoretically and measurements that were popping up. Once quantum mechanics was elucidated in its basic structure, then there were a bunch of experiments that were obvious to do, and people did those. People also just ran with the theory and predicted some things that couldn't be measured for decades. So who knows if they are going to be true or not? That's the interesting interplay between theory and experiment in that particular human intellectual endeavor.

Consider gravity for a moment. More than one person has made the argument to me that, since we know how gravity works, pretty soon we'll have an electronics that does the same thing with gravity that electronics today does with electrons and photons. But they skip the step that there is no quantum gravity. There is no theory yet that can be tested that says anything about quantum gravity, and that's in part because we don't have the measurement capability yet. No one has actually seen the graviton, the gravitational equivalent of the photon. It was only in the late '80s that gravitational radiation was shown to exist, but Einstein predicted that long ago. Our expectation is that just as with the wave and particle formulations of electromagnetism, the wave theory of gravity will be supplemented by something that involves particles. Thus the graviton. But there is no measurement that says anything about gravitons yet. As you think about the transition to electronics as a technology, despite everything people say about Moore's Law and running out of space with transistors, etc., we have explored only a really tiny corner of the technologies that are possible as a result of quantum mechanics; just as applied to electrons and protons and atoms, solid state physics, and that sort of thing. We are just beginning to investigate quantum computing and quantum cryptography. Think what's going to happen to technology when we get quantum gravity, as an analogy.

There are lots of people who have written science fiction along those lines—physicists, for example, like David Brin. That's the basis for his novel *Earth*—the development of what he called cavatronics. It's a technological manipulation of gravity. He suggested that a theorist finally made the connection and understood how gravity really worked. I don't remember whether Brin calls it "quantum gravity," but that's the language that we are using for it today. People are struggling with quantum gravity. There are all kinds of theory papers claiming this or that.

What's more, anytime you have a technology that allows you to measure something it is also, in principle, a technology for manipulating that thing. This is especially true in quantum mechanics—if you are interacting with the system, you are changing it. The definition of an interaction in quantum mechanics is a measurement.

GBN: Is that always true? If you have a technology that allows you to measure it, you can, by definition, manipulate it?

Carlson: At least in quantum mechanics. If there's a signal out there and you can measure the signal, then you have a way to interact with whatever the medium is that's changing. Similarly, getting back to biology, if you have a way to measure anything about a protein, you are, by definition, manipulating the protein in some way. It's not always the case that it's the best way to manipulate it, but it opens the door.

GBN: Schwartz also makes the argument that physics, chemistry and biology are on the verge of a conceptual revolution. You're saying that it's more like this yin/yang interaction between the two, theory and experiment.

Carlson: I tried to make a similar point about biology earlier. It's easy to spend time writing and exploring theories of physical systems, be they biological or solid state physics or chemical, when they're not constrained by data. Look at the number of theory papers that have turned out wrong compared to the number that were right. Or the number that make a really clean prediction of something, rather than just saying, "I was at the chalkboard and I phrased the question a little differently than anyone else and it led me down this road, and I did this math and here's the result I got." There is a lot of theoretical biology where people say, "I made this computer model and it had these characteristics, and my simulation results"—which they often call experimental data—"compared very well with my theory, on which the computer program was based." That's kind of a waste of time, in my opinion.

I think that understanding is based on our ability to predict and then to confirm the predictions, or not, based on measurement. That measurement often comes about from the development of a new tool. There are some very nice tools coming along now, and many of them will revolutionize the way we interact with various physical systems. That's where I see myself in science, as a tool builder. I've spent a long time thinking about how I wanted to model biology, about building models of neurobiological systems in particular. I've thought a lot about how information flows in biological systems and how information flows from the environment into the genome. But in the end I realized I just couldn't measure any of the things that I wanted to include as parameters in my models. I spend my time now thinking about how to better measure things.

Generational differences; young physicists going into biology

GBN: How old are you?

Carlson: I'm 31.

GBN: Do you see any generational differences in the approach to biology?

Carlson: Definitely. Let's look at some specific examples. There has been a spate of papers in the last few years in *Science* and *Nature* and publications like that where people are measuring properties of single molecules—single enzymes walking along DNA, or looking at the mechanical properties of DNA. If you look at who has published those papers, it's in general physicists or engineers who realized they could take a piece of equipment off of the physics shelf and slap a biological system in it and see what happened. That's not always the case—a lot of that work has biologists on the author list as well—but the impetus for a lot of the measurements and tool use comes from physics.

In like manner, recently there have been a small but growing number of papers by people who are trying to build quantitative models on a molecular scale. That is, you have one species of molecule doing this thing and another species of molecule doing another thing. We're going to try to write down as much of the physics as we think is important and keep track of what is going on and try to make a prediction. If you ask who's doing that, you'll find that it's physicists and engineers. If you ask who's really trying to build new things out of biology, how to really understand biological systems in a way that allows you to build something new out of them, then it's engineers and, now to a lesser extent, physicists.

It's certainly true that biologists have always built new things out of biology, in the sense that they've been shuffling genes around for decades now and they've been using enzymes that they find in nature to do things in the lab. They have also been changing those enzymes to suit their needs. The DNA polymerase enzyme used now to copy DNA in the lab is not a naturally occurring polymerase enzyme. It's a recombined enzyme that's been adjusted to work better under laboratory conditions. Biologists have built stuff all along the way. But as far as making biology understandable in a quantitative way, it's not generally biologists.

And the reason, I think, is that the tools are finally there that dispel the nonsense of making models with freedom from data. You can actually measure things now. Which people know how those tools work? Physicists and engineers. It's also true that a lot of that work is done by younger people, in part because they haven't seen the failures of the past. They don't know that it won't work.

Physicists have entered biology a couple of times over the years. A big effort by physicists in biology started molecular biology. It was Max Delbruck, etc., in Cambridge, starting a way of thinking about and doing biology that spread rapidly and was very successful. But they used techniques from physics. It wasn't really physics in the sense of fundamental physical phenomenon they were studying. It was statistics and such, with which physicists were very facile, and biologists weren't as ready to use. There are always biologists who are good at math, but not all of them are. A lot of biology programs don't require any physics at all, generally require very little math, and if you listen to Roger Brent, our Director at the Molecular Sciences Institute, talk about it, the biology curriculum now selects for people who are poor at that kind of thinking, and development of those abilities is not encouraged.

When I first got interested in biology, I was very small. My first memories from the age of 3 or 4 are of poking around in tide pools on the Oregon coast, wondering what was there and trying to figure out how it all worked together and how it all survived in that little space. In high school I didn't really like biology because it was just a list of facts, and that didn't help me understand what was going on. Then I wandered off to physics because I wanted to know how everything worked. By the time I hit grad school I realized that of course I could use the things I learned in physics to better understand biological systems.

We have in fact learned a great deal of physics from studying biology. We've learned about signal processing, about electron transfer on the molecular level. Protein folding is a condensed matter physics problem in itself that has yet to be really solved, though people are making good progress these days. Biology is not easy. It's complicated, it's messy, and you have to be careful.

One of the reasons that physicists have found it difficult to work on biology is that you really do have to know a lot of system-specific details. You have to be careful about the pH of the buffer, or the salt content of the buffer. You have to know that when you pipette—when you transfer liquids around—you have to agitate it just enough, but only so much so that you don't shear the DNA or denature the proteins or introduce air bubbles. There is a huge amount of stuff you have to know in order to get it right. It's art, by which I mean practiced laboratory skills, some of which take years to learn. So that accumulation of knowledge and the accumulation of details in your head about how it all works and the accumulation of experiments that you and other people have done has been the measure of success in biology. That's how you get tenure. That's how you get to be a big guy. Not only by doing good stuff, but also by standing up at conferences and saying, "That result is wrong, because I can see that you did this fundamental thing at the bench wrong. How stupid of you." That kind of thing, as far as the biology is concerned, tends to send physicists headed in the other direction.

The next big entry of physicists into biology was the late 1960s when new tools were being developed to study neurons. People developed tools like patch clamps to record electrical signals from individual cells and manipulate the electrical behavior of individual neurons. This was a good time for physicists because they could measure something again. "These quantitative models we have for how the brain works, we can test them. Not only that, when new experiments come along, we can make new theories and we're happy again." At the same time a lot of them said, "In 10 years we'll solve the brain." Ten years went by and clearly they hadn't solved the brain, so once again a lot of them went away with their tail between their legs.

So biologists said, “That is enough. Physicists can’t do biology.” They were right, to some extent. But enough time has passed that people my age and even older didn’t live through that, didn’t see it happen. And our professors were young enough that they didn’t have such a bad feeling about trying to do biology. Even so, while biological physics is kind of a hip thing in many physics departments, it’s not obvious whether it’s hip because they think it’s a good thing or because that’s where they can get money. There are a whole lot of physics professors out there who don’t get it at all—”It’s not real physics; it’s not interesting. We’re not going to learn any physics from biology, so why call it physics anyway?”

GBN: So it is hip within your generation?

Carlson: Oh, it’s very hip. And within the next generation as well. There are large numbers of students going down the road between biology and physics and learning things from both sides.

GBN: This is like the 1960’s wave—are you more confident that this group is not going to walk away again with their tails between their legs?

Carlson: At least not for a while. I think the reason is because the measurement techniques are finally there at the detail that physicists are taught to think about systems. Again, until recently the norm has been natural language models in biology. You can make a model of a black box running around out in the field, and there’s some kind of description you can attach to what the black box is doing. You can record that the black box eats green berries but doesn’t eat red berries and it cowers under a tree when a big bird flies by. That’s one kind of story you can tell, but that’s not very interesting to most physicists unless there are 100 black boxes running around and if they interact with a particular strength. Then I can make an abstract model about the behavior of black boxes, especially when I put in the green boxes, which might be the birds and the food, and I can make it complex and interesting to think about mathematically. Even models at that level are difficult to measure with detail that satisfies someone who is drawn to physics.

However, I should say that this is the sort of approach taken in population ecology and it has certainly made great strides, starting with Robert May and Edward O. Wilson and island biogeography. One of the reasons it’s successful and useful is that you can go out in the world and measure stuff. It’s not just this model of organisms that’s abstracted to differential equations, but you can go to an island with butterflies or lizards of a certain type, and because it’s an island, you can count them. You can’t do that very well on the mainland unless you find an isolated population.

But getting back to molecules, now it’s getting to the point where you can apply all of the stuff you learned in condensed matter physics to biology. It becomes useful, not just in a mathematical sense, to think about how polymers like DNA behave under certain circumstances because it’s not just on the blackboard anymore. It’s not just an idealized polymer in a solution. I can now use these conceptual tools to study the nucleus of a cell and I can compare my hypotheses to measurements. I can begin to understand why DNA is packaged in a certain way. I can explain to some extent, based on the physics of the DNA polymer, why an enzyme that manipulates DNA works in a particular way. In fact, I can infer, based on the behavior and packing of the DNA, that there must be an enzyme with a particular function even though no one has discovered it yet. That’s the kind of insight that a physicist wants but until recently has been difficult to produce on a regular basis.

There were certainly always people out there making interesting observations and measuring interesting things. But because of the increasingly broad distribution of sophisticated tools there is a growing diversity of people now pushing in really interesting directions. There is no paucity of problems that can be approached in this way.

GBN: That would be a good way to frame the coming decade, in a way. At the end of the decade are you pulling out your hair and running from the field?

Carlson: Having made that optimistic spiel about how things are going, I'm once again in a frame of mind where I'll say, "In 10 years we'll be able to design organisms and change people's hair color or grow feathers on their chin." We can do remarkable things today that we couldn't do even a few years ago. When I look at the work that is going on here and the way my thinking changes every day on how to solve problems, I'm actually incredibly optimistic. I suppose one caveat to consider is that everyone thinks they live in a special time. It doesn't matter who you are or when you live, you think that now is *the* time. Whether it's the end or the beginning of the world, the latest greatest revolution in whatever, you think that you're living in the most interesting time. Nevertheless, I really do think this is an amazing time.

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